

SOC Investigation Lab Report

Network Security Assessment Using Kali Linux, Nmap, Wireshark and Metasploitable 2



Prepared by: Samadhi Madubhashini

Date: July 2026

Table Of Content

Introduction.....	3
Project Objectives	3
Lab Environment	4
Lab Architecture.....	4
Tools Used	5
Methodology.....	5
Practical Activities.....	6
Results and Findings	11
Security Recommendations.....	14
Challenges Faced	14
Skills Gained	15
Conclusion	15
References	16

Introduction

A Security Operations Center (SOC) is responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity threats within an organization. SOC analysts use specialized tools to identify suspicious network activity, investigate security incidents, assess vulnerabilities, and recommend appropriate security controls.

This project was conducted to simulate a basic SOC investigation in a controlled virtual laboratory environment. Kali Linux was used as the analyst workstation, while Metasploitable 2 was used as the intentionally vulnerable target machine. VMware Workstation provided the isolated virtual network that allowed the investigation to be performed safely without affecting production systems.

Throughout the project, network reconnaissance, service enumeration, vulnerability assessment, packet capture, and basic system investigation were carried out using industry-recognized security tools. The findings were analyzed from a SOC analyst's perspective to identify security weaknesses and recommend practical mitigation strategies.

Project Objectives

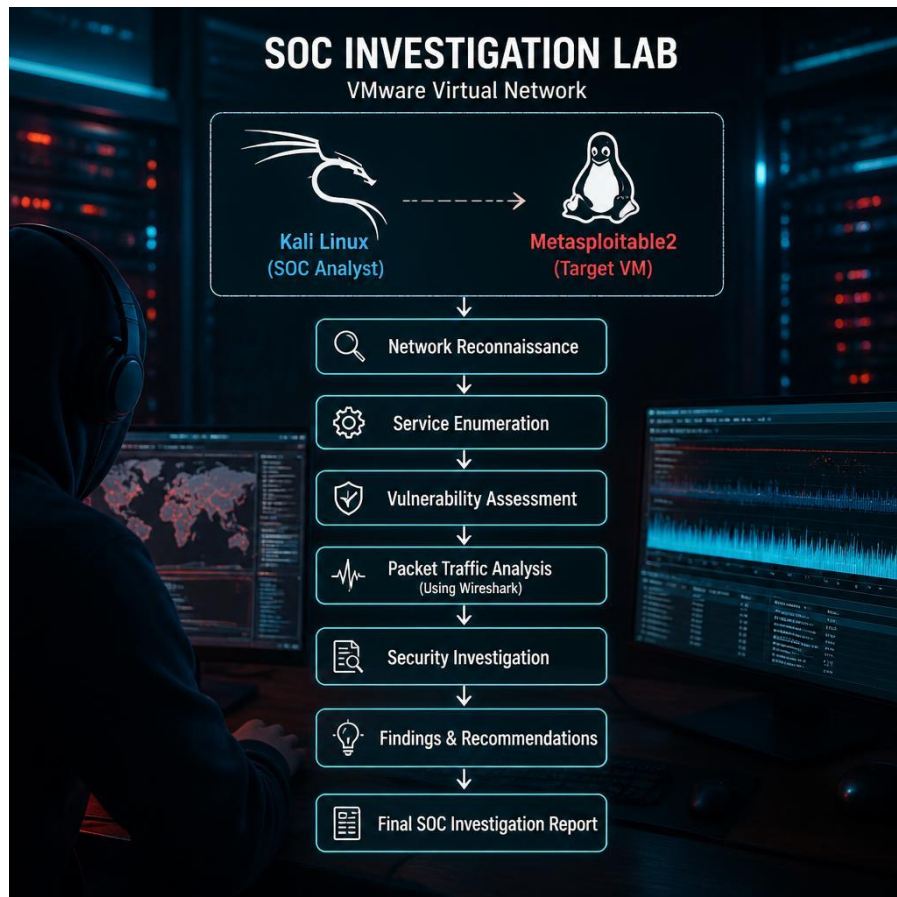
The objectives of this project were:

1. Design a safe virtual laboratory for cybersecurity investigations.
2. Verify network connectivity between the analyst and target systems.
3. Perform host discovery and service enumeration using Nmap.
4. Identify operating system information and exposed network services.
5. Conduct vulnerability assessment using Nmap NSE scripts.
6. Capture and analyze network traffic using Wireshark.
7. Investigate selected services such as FTP and SSH.
8. Identify security risks and document findings from a SOC analyst's perspective.
9. Recommend security improvements based on the investigation results.

Lab Environment

Component	Description
Analyst Machine	Kali Linux
Target Machine	Metasploitable 2
Virtualization Platform	VMware Workstation
Network Type	VMware Virtual Network
Scanning Tool	Nmap
Packet Analysis Tool	Wireshark
Investigation Tools	SSH, FTP
Operating System	Linux

Lab Architecture



Tools Used

Tool	Purpose
VMware Workstation	Virtual environment
Kali Linux	Security analyst workstation
Metasploitable 2	Vulnerable target system
Nmap	Network scanning and service detection
Wireshark	Network packet capture and analysis
FTP	Service investigation
SSH	Remote administration and investigation

Methodology

The investigation followed a structured workflow commonly used in Security Operations Centers.

Step 1 – Network Connectivity Verification (ping): Connectivity between Kali Linux and Metasploitable 2 was verified using the ping command to ensure both systems could communicate across the virtual network.

Step 2 – Service Enumeration: Nmap service version detection (nmap -sV) was used to identify open ports and determine the services running on the target machine.

Step 3 – Operating System Detection: Operating system detection (nmap -O) was performed to identify the target platform and gather additional reconnaissance information.

Step 4 – Vulnerability Assessment: Nmap NSE vulnerability scripts (nmap --script vuln) were executed to identify known weaknesses associated with exposed services.

Step 5 – Network Traffic Monitoring: Wireshark captured the network traffic generated during scanning activities. Display filters such as icmp, tcp, and TCP flag filters were applied to analyze reconnaissance traffic.

Step 6 – Service Investigation: Individual services, including FTP and SSH, were examined to validate the Nmap findings and identify security weaknesses.

Step 7 – Risk Assessment: The collected evidence was analyzed to identify security risks, evaluate their impact, and recommend appropriate mitigation measures.

Practical Activities

Activity	Status	Evidence
VMware Lab Setup	✓ Completed	Figure 1 – VMware virtual machines
Network Connectivity Test	✓ Completed	Figure 2 – Successful ping test
Nmap Service Scan	✓ Completed	Figure 3 – Nmap -sV scan output
Operating System Detection	✓ Completed	Figure 4 – Nmap -O scan result
Vulnerability Scan	✓ Completed	Figure 5 – Nmap NSE vulnerability scan
Wireshark Packet Capture	✓ Completed	Figure 6 – Packet capture interface
FTP Investigation	✓ Completed	Figure 7 – Anonymous FTP login
SSH Investigation	✓ Completed	Figure 8 – Successful SSH login
Service Verification	✓ Completed	Figure 9 – netstat -tulnp output

Practical Evidence

Figure 1: VMware Workstation showing Kali Linux and Metasploitable 2 virtual machines.

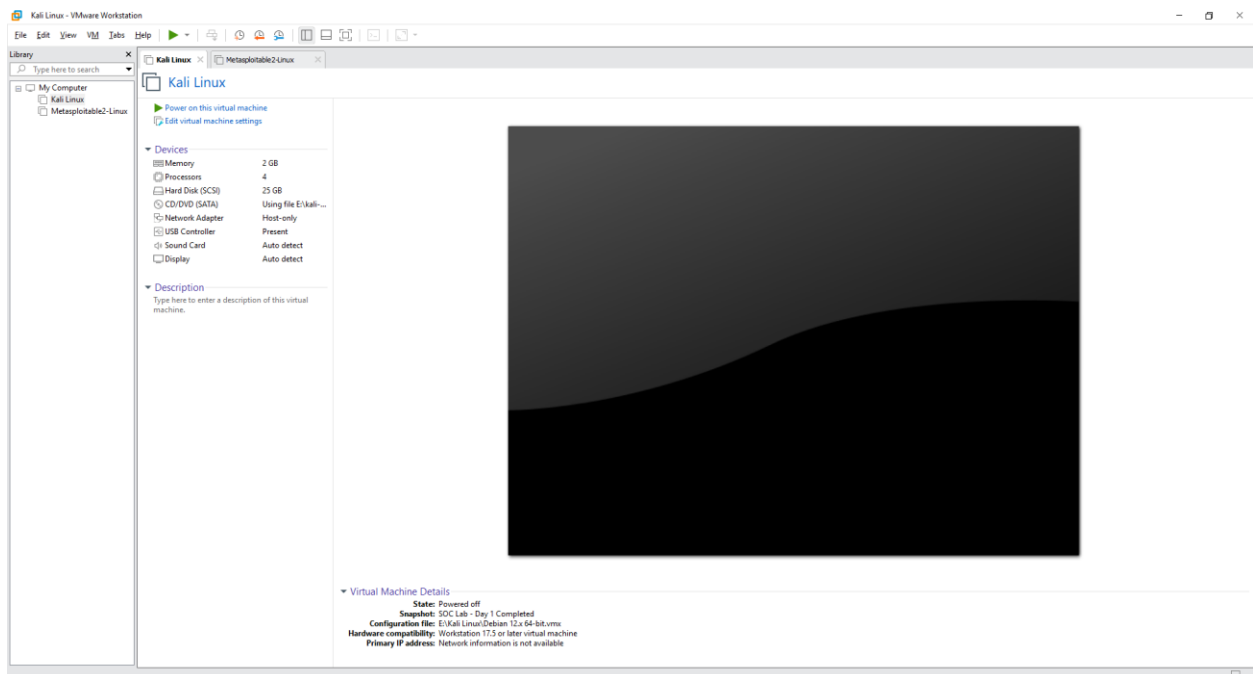
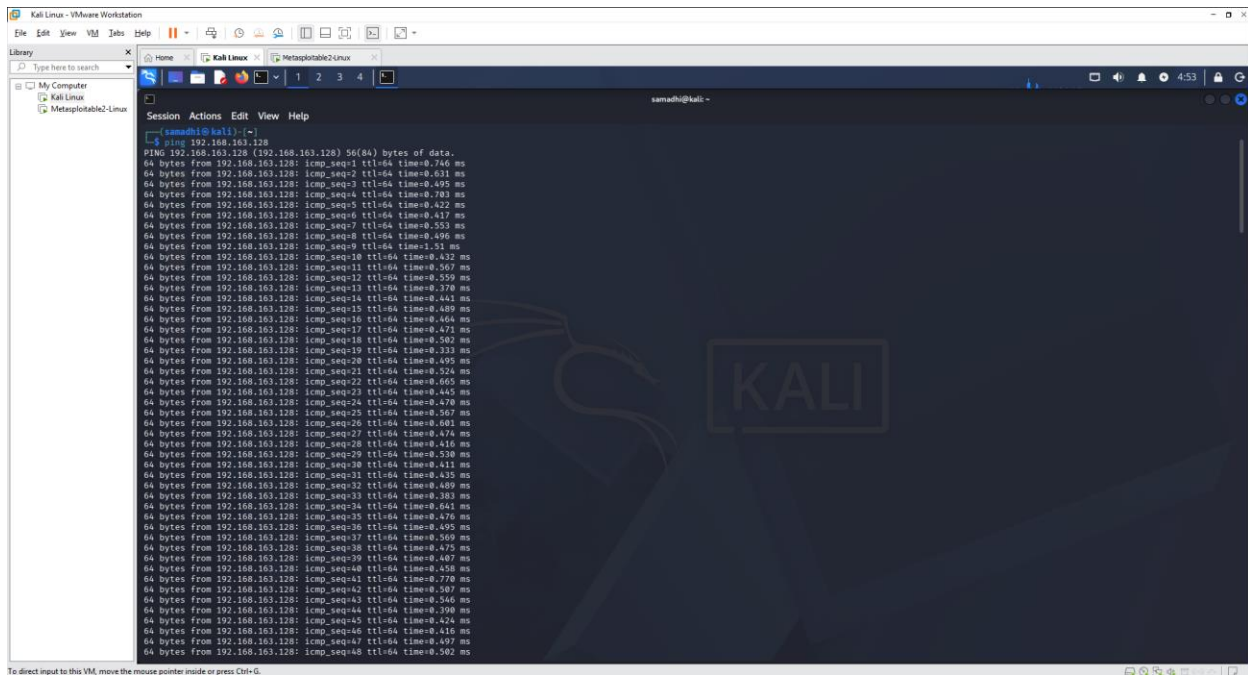
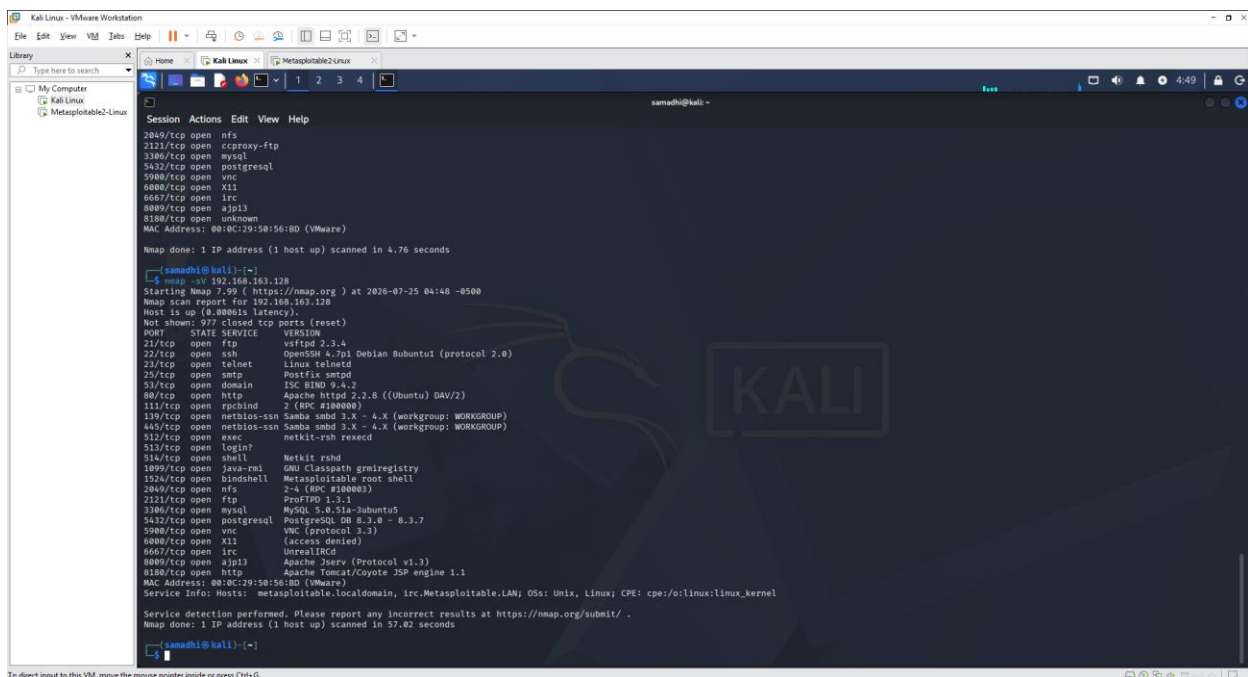


Figure 2: Successful network connectivity test using the ping command.



```
[samadhi@kali:~]$ ping 192.168.163.128
PING 192.168.163.128 (192.168.163.128) 56(84) bytes of data.
64 bytes from 192.168.163.128: icmp_seq=1 ttl=64 time=0.746 ms
64 bytes from 192.168.163.128: icmp_seq=2 ttl=64 time=0.631 ms
64 bytes from 192.168.163.128: icmp_seq=3 ttl=64 time=0.495 ms
64 bytes from 192.168.163.128: icmp_seq=4 ttl=64 time=0.703 ms
64 bytes from 192.168.163.128: icmp_seq=5 ttl=64 time=0.422 ms
64 bytes from 192.168.163.128: icmp_seq=6 ttl=64 time=0.417 ms
64 bytes from 192.168.163.128: icmp_seq=7 ttl=64 time=0.553 ms
64 bytes from 192.168.163.128: icmp_seq=8 ttl=64 time=0.496 ms
64 bytes from 192.168.163.128: icmp_seq=9 ttl=64 time=0.511 ms
64 bytes from 192.168.163.128: icmp_seq=10 ttl=64 time=0.432 ms
64 bytes from 192.168.163.128: icmp_seq=11 ttl=64 time=0.567 ms
64 bytes from 192.168.163.128: icmp_seq=12 ttl=64 time=0.559 ms
64 bytes from 192.168.163.128: icmp_seq=13 ttl=64 time=0.378 ms
64 bytes from 192.168.163.128: icmp_seq=14 ttl=64 time=0.441 ms
64 bytes from 192.168.163.128: icmp_seq=15 ttl=64 time=0.489 ms
64 bytes from 192.168.163.128: icmp_seq=16 ttl=64 time=0.464 ms
64 bytes from 192.168.163.128: icmp_seq=17 ttl=64 time=0.471 ms
64 bytes from 192.168.163.128: icmp_seq=18 ttl=64 time=0.502 ms
64 bytes from 192.168.163.128: icmp_seq=19 ttl=64 time=0.333 ms
64 bytes from 192.168.163.128: icmp_seq=20 ttl=64 time=0.495 ms
64 bytes from 192.168.163.128: icmp_seq=21 ttl=64 time=0.524 ms
64 bytes from 192.168.163.128: icmp_seq=22 ttl=64 time=0.665 ms
64 bytes from 192.168.163.128: icmp_seq=23 ttl=64 time=0.465 ms
64 bytes from 192.168.163.128: icmp_seq=24 ttl=64 time=0.478 ms
64 bytes from 192.168.163.128: icmp_seq=25 ttl=64 time=0.567 ms
64 bytes from 192.168.163.128: icmp_seq=26 ttl=64 time=0.601 ms
64 bytes from 192.168.163.128: icmp_seq=27 ttl=64 time=0.474 ms
64 bytes from 192.168.163.128: icmp_seq=28 ttl=64 time=0.416 ms
64 bytes from 192.168.163.128: icmp_seq=29 ttl=64 time=0.538 ms
64 bytes from 192.168.163.128: icmp_seq=30 ttl=64 time=0.411 ms
64 bytes from 192.168.163.128: icmp_seq=31 ttl=64 time=0.435 ms
64 bytes from 192.168.163.128: icmp_seq=32 ttl=64 time=0.409 ms
64 bytes from 192.168.163.128: icmp_seq=33 ttl=64 time=0.383 ms
64 bytes from 192.168.163.128: icmp_seq=34 ttl=64 time=0.661 ms
64 bytes from 192.168.163.128: icmp_seq=35 ttl=64 time=0.476 ms
64 bytes from 192.168.163.128: icmp_seq=36 ttl=64 time=0.495 ms
64 bytes from 192.168.163.128: icmp_seq=37 ttl=64 time=0.369 ms
64 bytes from 192.168.163.128: icmp_seq=38 ttl=64 time=0.475 ms
64 bytes from 192.168.163.128: icmp_seq=39 ttl=64 time=0.407 ms
64 bytes from 192.168.163.128: icmp_seq=40 ttl=64 time=0.458 ms
64 bytes from 192.168.163.128: icmp_seq=41 ttl=64 time=0.778 ms
64 bytes from 192.168.163.128: icmp_seq=42 ttl=64 time=0.507 ms
64 bytes from 192.168.163.128: icmp_seq=43 ttl=64 time=0.346 ms
64 bytes from 192.168.163.128: icmp_seq=44 ttl=64 time=0.398 ms
64 bytes from 192.168.163.128: icmp_seq=45 ttl=64 time=0.424 ms
64 bytes from 192.168.163.128: icmp_seq=46 ttl=64 time=0.415 ms
64 bytes from 192.168.163.128: icmp_seq=47 ttl=64 time=0.497 ms
64 bytes from 192.168.163.128: icmp_seq=48 ttl=64 time=0.502 ms
```

Figure 3: Nmap service version detection (nmap -sV) identifying open ports and running services.



```
[samadhi@kali:~]$ nmap -sV 192.168.163.128
Starting Nmap 7.90 ( https://nmap.org ) at 2020-07-25 04:48 -0500
Nmap scan report for 192.168.163.128
Host is up (0.00001s latency).
Not shown: 971 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
22/tcp    open  ftp             vsftpd 2.3.4
22/tcp    open  ssh             OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet          Linux telnetd
25/tcp    open  smtp            Postfix smtpd
53/tcp    open  domain          ISC BIND 9.4.2
80/tcp    open  http            Apache httpd 2.2.0 ((Ubuntu) DAV/2)
111/tcp   open  rsh             2 (RC #100000)
139/tcp   open  netbios-ssn     Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn     Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec            netkit-rsh execd
513/tcp   open  login?
514/tcp   open  shell           Netkit rshd
1099/tcp  open  java-remote     GNU Classpath gmrregistry
1524/tcp  open  bindshell       Metasploitable root shell
2049/tcp  open  nfs             2-4 (RPC #100003)
2121/tcp  open  ftp             ProFTPD 1.3.5-1
3306/tcp  open  mysql           MySQL 5.0.51a-Jubuntu5
5432/tcp  open  postgresql      PostgreSQL DB 8.3.8 - 8.3.7
5900/tcp  open  vnc             VNC (protocol 3.3)
6000/tcp  open  X11             (access denied)
6067/tcp  open  irc             UnrealIRCd
8009/tcp  open  ajp13           Apache Jkery (Protocol v1.3)
8188/tcp  open  http            Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:0C:29:15:05:BD (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 57.02 seconds
[samadhi@kali:~]$
```


Figure 4: Operating system detection using nmap -O.

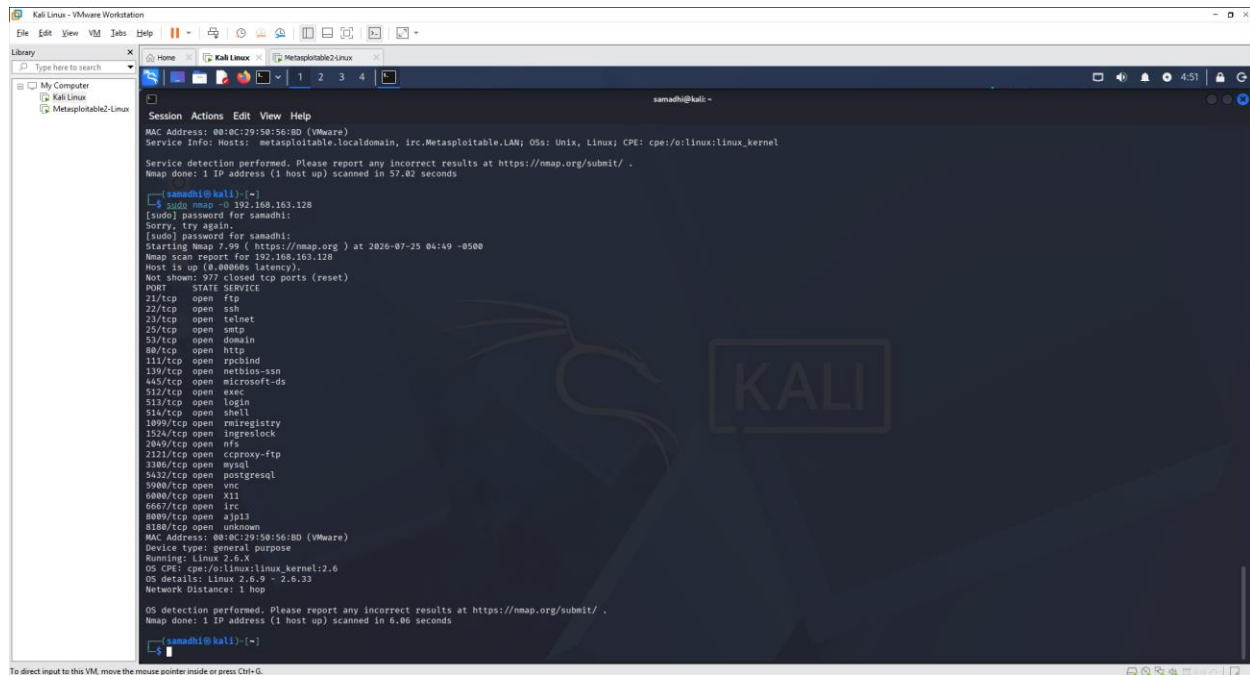


Figure 5: Vulnerability assessment using Nmap NSE scripts (nmap --script vuln).

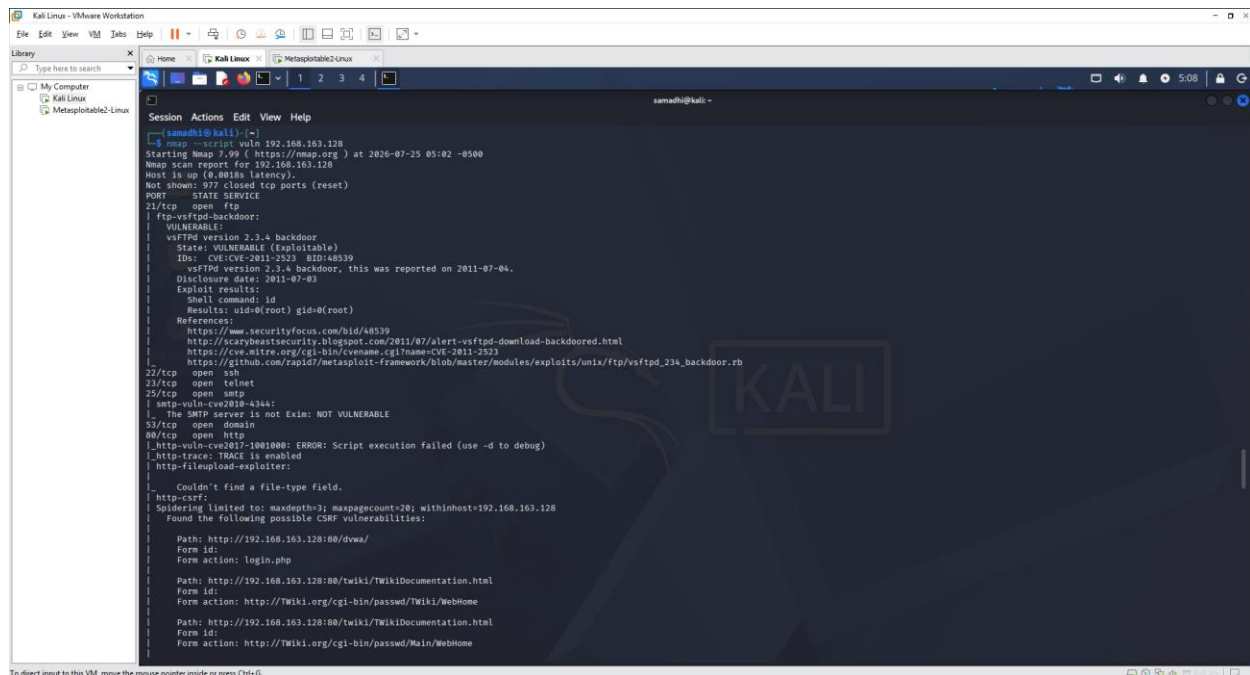
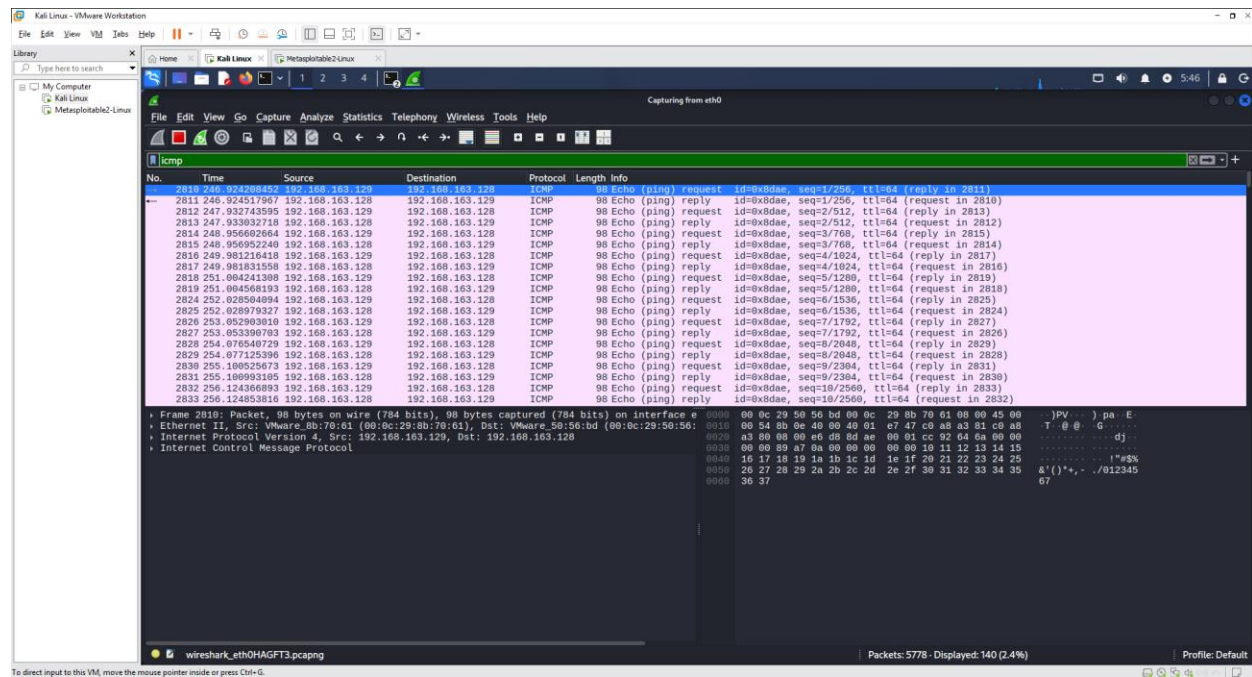


Figure 6: Wireshark packet capture displaying ICMP and TCP traffic generated during the investigation.

6.1 - ICMP: Internet Control Message Protocol



6.2 - TCP: Transmission Control Protocol

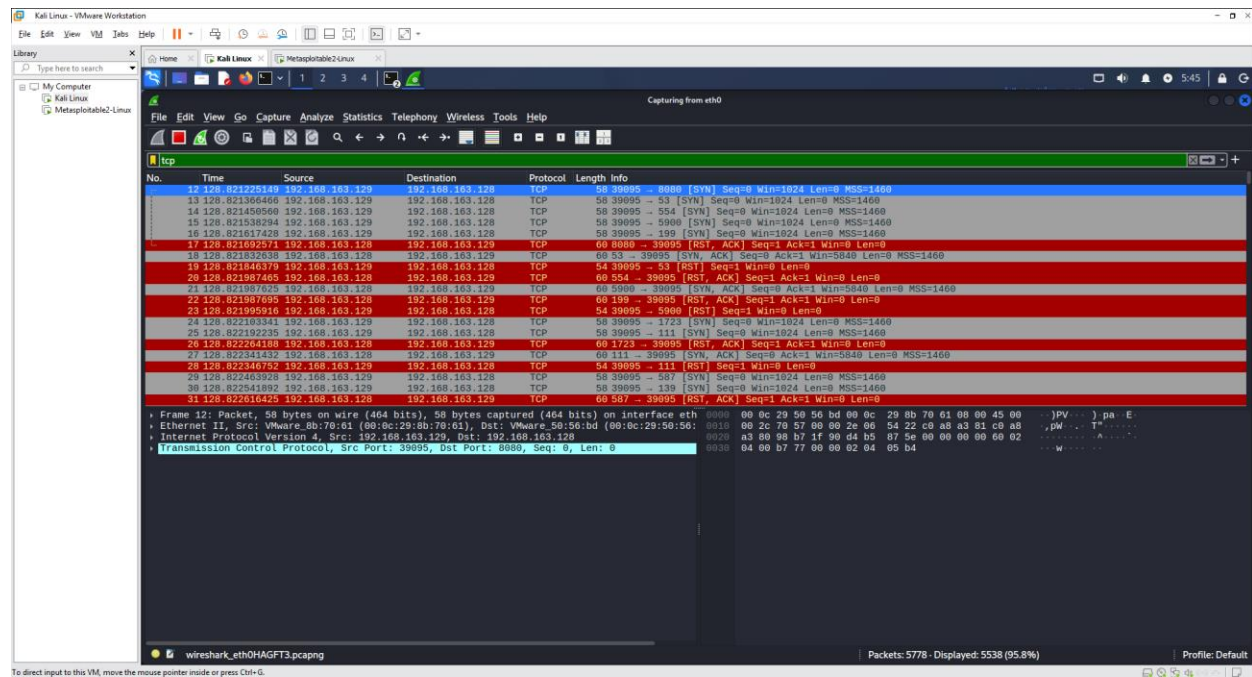
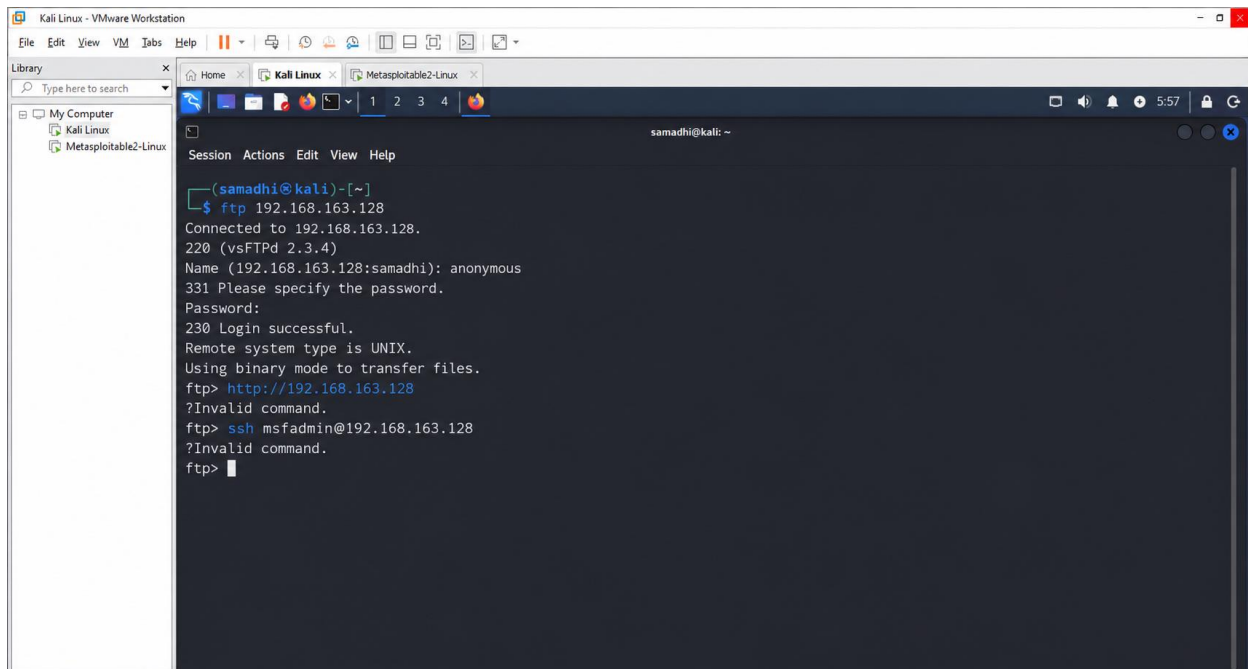


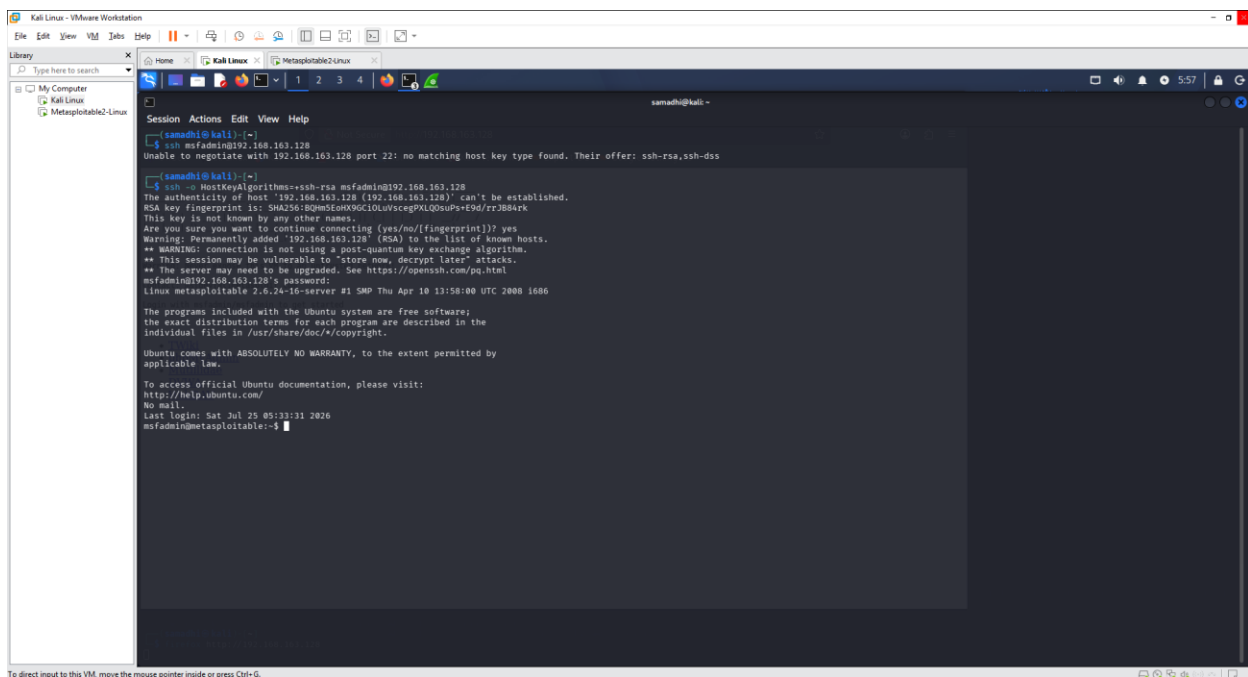
Figure 7: Successful anonymous FTP login to the target machine.



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali Linux
Metasploitable2-Linux

samadhi@kali: ~
Session Actions Edit View Help
(samadhi@kali)-[~]
$ ftp 192.168.163.128
Connected to 192.168.163.128.
220 (vsFTPD 2.3.4)
Name (192.168.163.128:samadhi): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> http://192.168.163.128
?Invalid command.
ftp> ssh msfadmin@192.168.163.128
?Invalid command.
ftp>
```

Figure 8: Successful SSH connection to Metasploitable 2 after enabling the required host key algorithm.



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Kali Linux
Metasploitable2-Linux

samadhi@kali: ~
Session Actions Edit View Help
(samadhi@kali)-[~]
$ ssh msfadmin@192.168.163.128
Unable to negotiate with 192.168.163.128 port 22: no matching host key type found. Their offer: ssh-rsa,ssh-dss

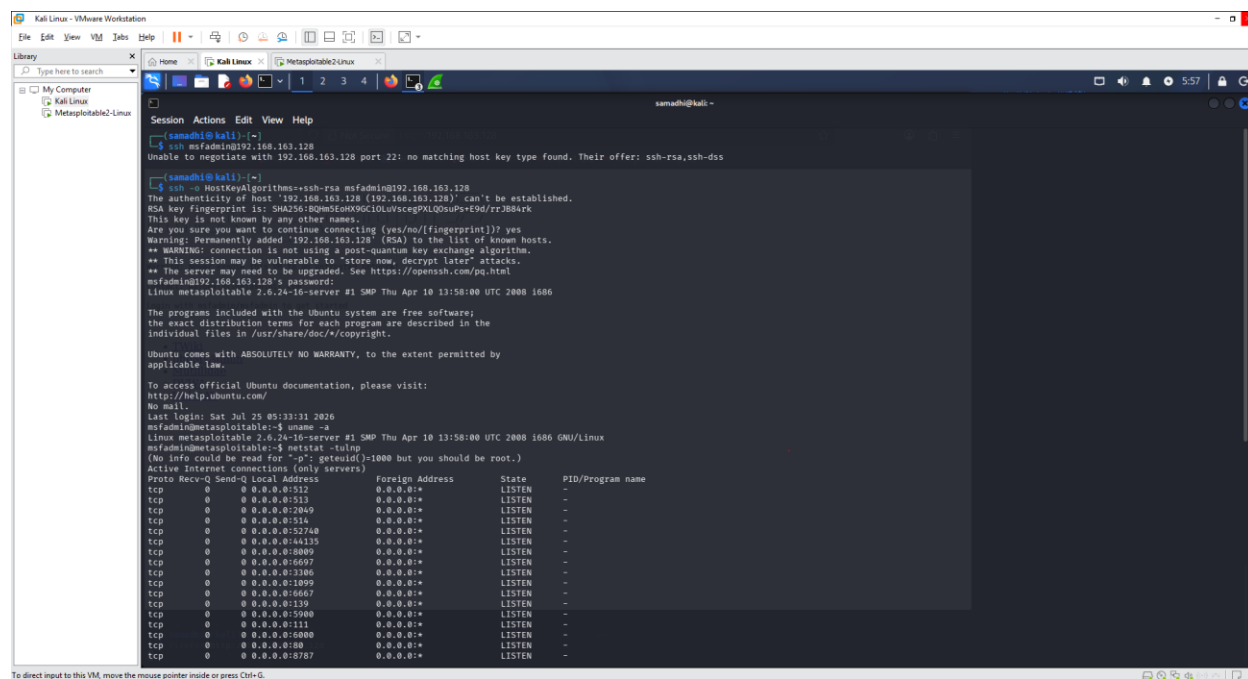
(samadhi@kali)-[~]
$ ssh -o HostKeyAlgorithms=ssh-rsa msfadmin@192.168.163.128
The authenticity of host '192.168.163.128 (192.168.163.128)' can't be established.
RSA key fingerprint is SHA256:8qumf8onx9CidUv5cep9ALQou8s+1Pd/rz3B4xk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.163.128' (ssh) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/04.html
msfadmin@192.168.163.128's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Sat Jul 25 05:33:31 2020
msfadmin@metasploitable:~$
```

Figure 9: Running services on the target machine verified using netstat -tulnp.



```
(samadhi@kali) ~$ ssh msfadmin@192.168.163.128
Unable to negotiate with 192.168.163.128 port 22: no matching host key type found. Their offer: ssh-rsa,ssh-dss

(samadhi@kali) ~$ ssh -o HostKeyAlgorithms=ssh-rsa msfadmin@192.168.163.128
The authenticity of host '192.168.163.128 (192.168.163.128)' can't be established.
RSA key fingerprint is: SHA256:BQW5e0XK9G10Luvscg9XL00suPs+9d/rR3884rk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.163.128' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
msfadmin@192.168.163.128 ~$ cat /etc/passwd
msfadmin@192.168.163.128 ~$ cat /etc/passwd
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Sat Jul 25 05:33:31 2026
msfadmin@metasploitable:~$ uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
msfadmin@metasploitable:~$ netstat -tulnp
(No info could be read for "-p": geteuid()=1000 but you should be root.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22             0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:5133           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:2849           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:514            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:52748          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:44135          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:8089           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6697           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:3386           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:1899           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6667           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:139            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:5080           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:111            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6000           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:80             0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:8787           0.0.0.0:*               LISTEN      -
```

Results and Findings

1. Network Connectivity Verification

Findings:

- The target machine was reachable through ICMP echo requests.
- Successful ping responses confirmed network connectivity between Kali Linux and the target system.

Reference: (Refer to Figure 2 – Successful ping test)

2. Nmap Network and Service Discovery Results

Findings:

- Nmap identified multiple open ports and running services on the target system. Exposed services increase the attack surface and may provide potential entry points for attackers.

Identified Services:

- FTP service detected
- SSH service detected

Reference: (Refer to Figure 3 – Nmap -sV Scan Output)

3. Anonymous FTP Login Vulnerability

Findings:

- The FTP service allowed anonymous authentication. This configuration allows users to access the FTP server without valid credentials, which may result in unauthorized file access and potential information disclosure.

Reference: (Refer to Figure 7 – Anonymous FTP Login)

4. Weak SSH Algorithms

Findings:

- The SSH service was accessible remotely. Security analysis identified weak or outdated SSH algorithms/ciphers, which may reduce the confidentiality and security of remote connections.
- Weak cryptographic configurations can increase the risk of downgrade attacks or exploitation of known vulnerabilities.

Reference: (Refer to Figure 8 – Successful SSH Login)

5. Vulnerability Scan Results

Findings:

The vulnerability assessment identified several security weaknesses, including:

- Outdated Linux kernel version
- Misconfigured services
- Potential vulnerabilities in installed applications

These weaknesses may increase the risk of system compromise if not properly patched or secured.

Reference: (Refer to Figure 5 – Nmap NSE Vulnerability Scan)

6. Wireshark Packet Analysis

Findings:

Wireshark packet analysis was performed to observe network communication behavior.

The captured traffic included:

1. ICMP Packets

- Generated during ping testing.
- Confirmed host availability.

Reference: (Refer to 6.1 - ICMP: Internet Control Message Protocol)

2. TCP SYN Packets

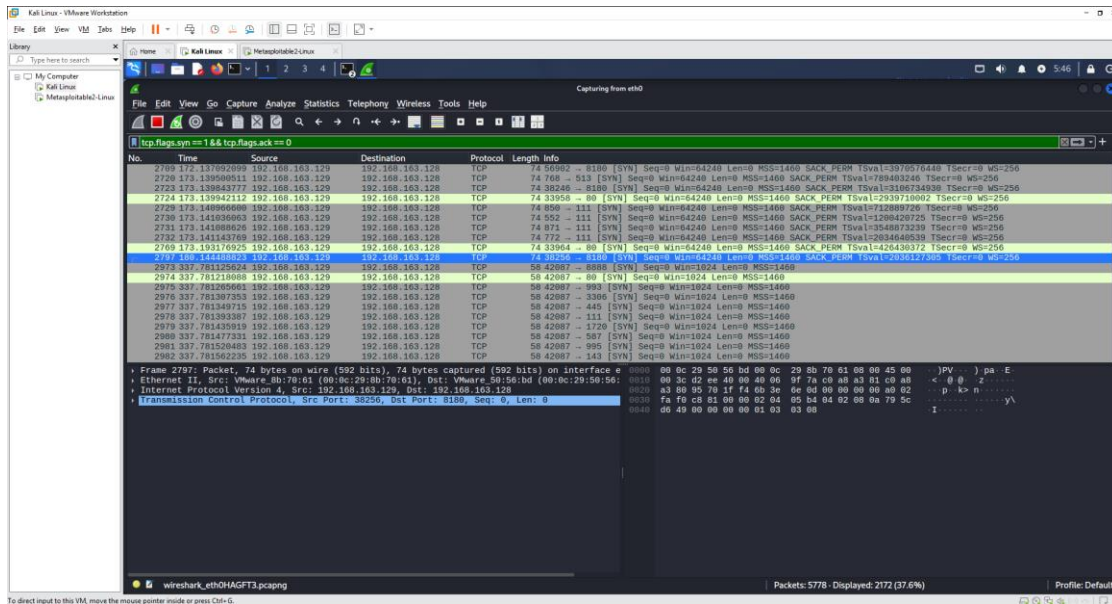
- Generated during connection attempts to services.

Reference: (Refer to 6.2 - TCP: Transmission Control Protocol)

3. SYN/ACK Packets

- Indicated that services were accepting TCP connection requests.

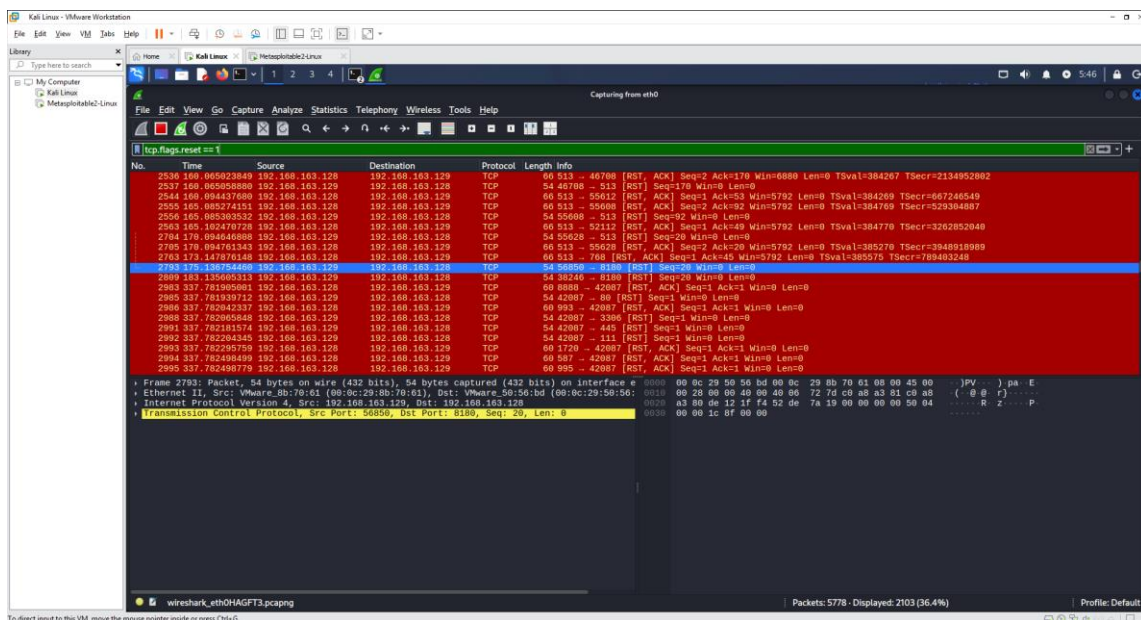
Reference:



4. RST Packets

- Indicated rejected or closed connections.

Reference:



Security Recommendations

The investigation identified several security improvements that would reduce the attack surface of the target system.

1. Disable anonymous FTP access unless there is a justified operational requirement.
2. Replace insecure services such as Telnet with encrypted alternatives such as SSH.
3. Upgrade the operating system and apply current security patches.
4. Update or replace outdated software versions that contain known vulnerabilities.
5. Disable unnecessary network services to minimize the attack surface.
6. Configure host-based and network firewalls to restrict unauthorized access.
7. Perform regular vulnerability assessments and penetration testing.
8. Continuously monitor network traffic and security logs to detect suspicious activity at an early stage.

Challenges Faced

During the practical assessment, several challenges were encountered while setting up the virtual lab, performing security tests, and analyzing results. These issues were resolved through troubleshooting and applying appropriate technical solutions.

1) Initial Network Connectivity Configuration

- **Challenge:** Initial communication between Kali Linux and the target machine was unsuccessful due to network configuration issues.
- **Resolution:** VMware network settings were adjusted, and connectivity was verified using ping tests before starting security assessments.

2) SSH Compatibility Issues with Modern OpenSSH

- **Challenge:** SSH connection issues occurred because of differences between older SSH configurations and modern OpenSSH security settings.
- **Resolution:** SSH settings were analyzed, and suitable connection methods were applied to successfully test the SSH service.

3) Understanding Wireshark Display Filters

- **Challenge:** Filtering and analyzing large amounts of captured network traffic was difficult initially.

- **Resolution:** Wireshark display filters such as `icmp`, `tcp.flags.syn==1`, and `tcp.flags.reset==1` were used to identify specific packet types.

4) Organizing Scan Results and Evidence

- **Challenge:** Managing outputs from Nmap, vulnerability scans, FTP, SSH, and Wireshark activities was challenging.
- **Resolution:** Results were organized by activity, and screenshots were labelled with figure numbers to create clear documentation.

5) Vulnerability Identification and Interpretation

- **Challenge:** Interpreting security issues such as exposed services, anonymous FTP access, weak SSH settings, and outdated components required additional analysis.
- **Resolution:** Each finding was reviewed to understand its security impact and documented with possible risks and improvements.

Skills Gained

Technical Skills	Professional Skills
Nmap (Network Scanning)	Analytical thinking
Wireshark (Packet Analysis)	Problem-solving
Linux Command Line	Technical Documentation
Network Analysis	Security Investigation
Vulnerability Assessment	Critical Thinking
Service Enumeration	Attention to Detail
Network Troubleshooting	Evidence Collection
SSH & FTP Investigation	Report Writing
Operating System Identification	Risk Assessment
Virtual Machine Management (VMware)	Communication of Technical Findings

Conclusion

This project successfully demonstrated the process of conducting a basic Security Operations Centre (SOC) investigation within a controlled virtual lab environment. By using Kali Linux, Metasploitable 2, Nmap, and Wireshark, activities such as network discovery, service enumeration, vulnerability assessment, packet analysis, and system investigation were performed successfully.

The assessment identified several security weaknesses, including anonymous FTP access, weak SSH configurations, exposed network services, and outdated system components. These findings were analyzed to understand their potential security impact, and appropriate recommendations were considered to improve system security.

Through this project, practical knowledge was gained in network security testing, Linux system analysis, vulnerability identification, packet inspection, evidence collection, and cybersecurity documentation. The experience provided valuable insight into the investigation techniques used by SOC analysts when monitoring, analyzing, and responding to security issues.

Overall, this project improved technical skills and strengthened the understanding of the importance of continuous monitoring, secure configuration, and vulnerability management in maintaining a secure IT environment.

References

1. Nmap Development Team. *Nmap Reference Guide: Official Documentation*. Available at: <https://nmap.org/book/man.html>
2. Wireshark Foundation. *Wireshark User Guide and Documentation*. Available at: <https://www.wireshark.org/docs/>
3. Rapid7. *Metasploitable 2 Documentation and Vulnerable Testing Environment*. Available at: <https://docs.rapid7.com/metasploit/>
4. VMware. *VMware Workstation User Documentation*. Available at: <https://docs.vmware.com/>
5. NIST. *Guide to Computer Security Log Management (SP 800-92)*. National Institute of Standards and Technology. Available at: <https://csrc.nist.gov/publications/detail/sp/800-92/final>
6. NIST. *Computer Security Incident Handling Guide (SP 800-61 Rev. 2)*. National Institute of Standards and Technology. Available at: <https://csrc.nist.gov/publications/detail/sp/800-61/rev-2/final>
7. Cisco Networking Academy. *Introduction to Cybersecurity and Network Security Learning Materials*. Available at: <https://www.netacad.com/>
8. TryHackMe. *Cybersecurity Training Labs and Learning Resources*. Available at: <https://tryhackme.com/>